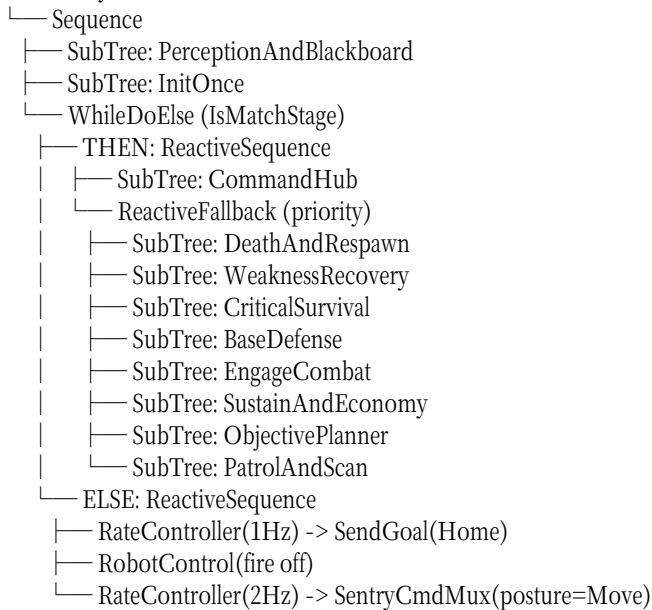


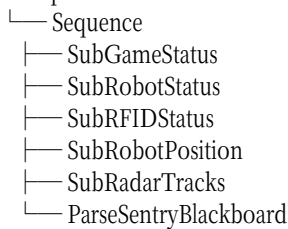
行为树树状图（缩进示意）

说明：该图是 SentryBehaviorTree.xml 的结构化缩进视图，用于快速对齐 Groot/BT.CPP 的节点层级。

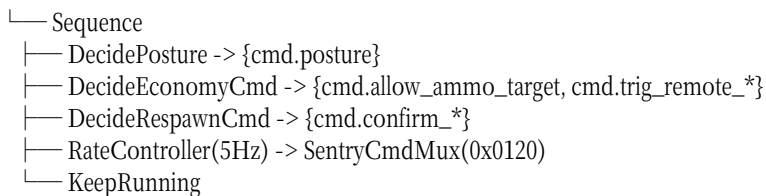
MainSentryTree



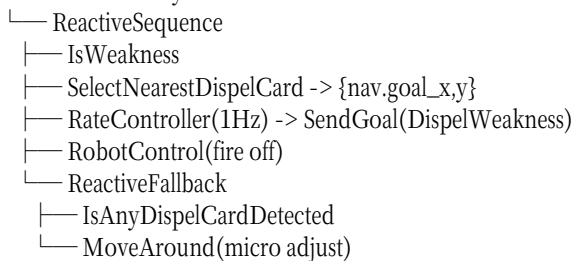
PerceptionAndBlackboard



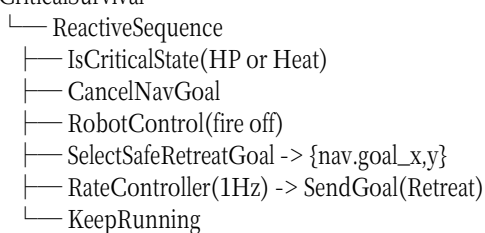
CommandHub



WeaknessRecovery



CriticalSurvival



BaseDefense

- └─ ReactiveSequence
 - └─ IsBaseThreatened
 - └─ RobotControl(fire on)
 - └─ RateController(1Hz) -> SendGoal(DefendAnchor)
 - └─ SubTree: CombatLoop

EngageCombat

- └─ ReactiveSequence
 - └─ HasValidTarget
 - └─ IsCombatAllowed
 - └─ RobotControl(spin, fire on)
 - └─ SubTree: CombatLoop

CombatLoop

- └─ ReactiveSequence
 - └─ SelectBestTarget -> {combat.best_target}
 - └─ AimAtTarget
 - └─ ReactiveFallback
 - └─ IsFireWindowOk
 - └─ RobotControl(fire off)
 - └─ FireBurst(burst/pause)
 - └─ KeepRunning

SustainAndEconomy

- └─ ReactiveFallback
 - └─ SubTree: HealPlan
 - └─ SubTree: AmmoPlan

HealPlan

- └─ ReactiveSequence
 - └─ IsHPBelow
 - └─ ReactiveFallback
 - └─ (At SUPPLY) -> HoldAndHeal
 - └─ GoSupply -> SendGoal + KeepRunning

AmmoPlan

- └─ ReactiveSequence
 - └─ IsAmmoBelow
 - └─ ReactiveFallback
 - └─ (At SUPPLY) -> HoldForSupplyAmmoTick
 - └─ SelectNearestResupplyStation -> SendGoal + KeepRunning

ObjectivePlanner

- └─ ReactiveSequence
 - └─ SelectObjective -> {nav.objective, nav.goal_x,y}
 - └─ RateController(1Hz) -> SendGoal(objective)
 - └─ RobotControl(fire off)
 - └─ (Arrive) IsAtGoal
 - └─ HoldObjective(hold_ms)

PatrolAndScan

- └─ Sequence
 - └─ RobotControl(scan, fire off)
 - └─ WaypointPatrol -> {nav.goal_x,y}
 - └─ RateController(1Hz) -> SendGoal(Patrol)
 - └─ KeepRunning